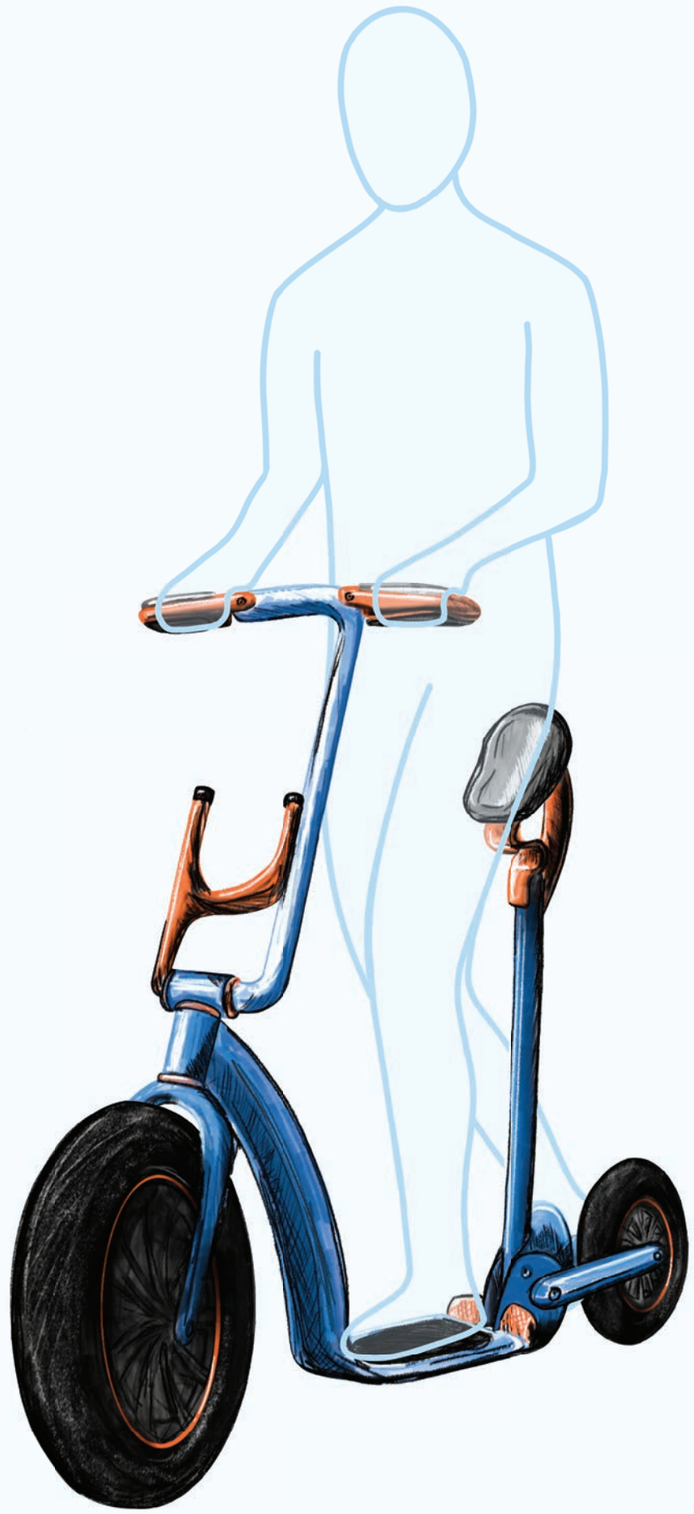
Motus

A solution for last mile commute



Concept

A single-person, portable mode of transport is needed to streamline urban commuting in Indian cities.

This solution aims to address the time-consuming aspects of walking to and from public transport stops. This solution should significantly reduce the time spent on walking portions of the journey.

Qualitative Requirements

- A personal vehicle that provides an option for last mile commute
- Should be portable
- Easy to carry when not riding
- Faster than walking

Test Rides

Using kick scooter to travel in Ahmadabad to understand the experience and to find problems and opportunities

While Riding

tired because of sun

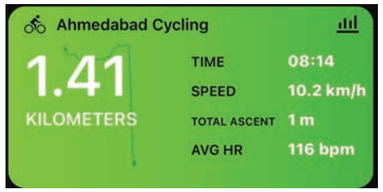
The hamstrings, Glutes, calf of the leg that is on the scooter get tired

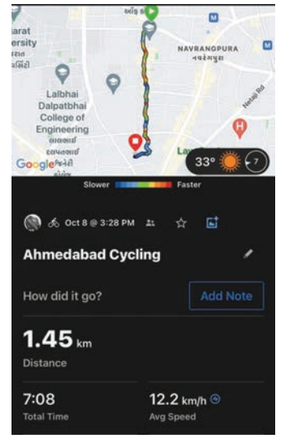
Cant lift using handle cause the body turns

Hard to interchange legs

Feels like you should go more distance in a kick → Momentum ?

Small bumps caused obstruction → The bump threshold !





Going in Metro

Have to fold while → A bit klinky while folding

Have to put in the bag → The reason? → should'nt hurt others?

Have to hold it in a hand → Not able to rest it on leg



While On Metro

Have to lift the whole

While sitting, where to → Not able to rest it on leg



Analysis

Strain on knee while kicking cause extra exertion



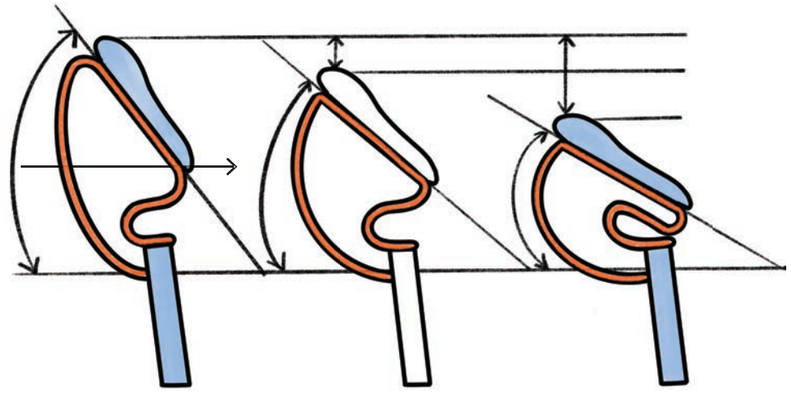
Strain on KNEE !!!

Solution

Down force of bodyweight is absorbed by complaint mechanism



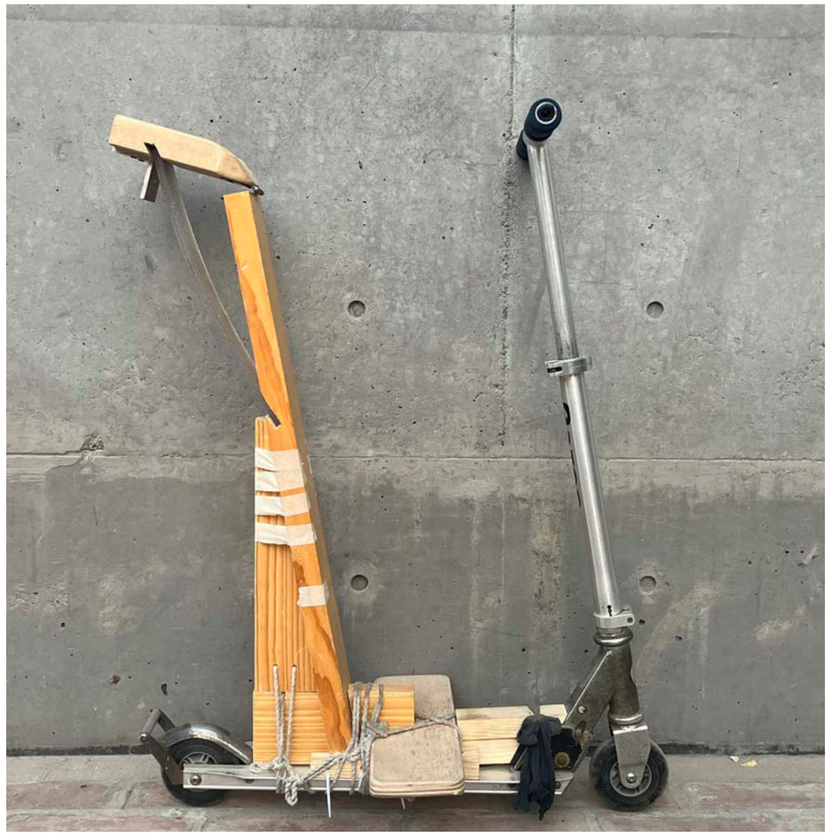
Bodyweight is taken by the seat



Progression of the seat spring

Proof of Concept

Testing a leaning element with a compliant mechanism by attaching it to an existing scooter



A leaning device with compliant mechanism attached to an existing scooter

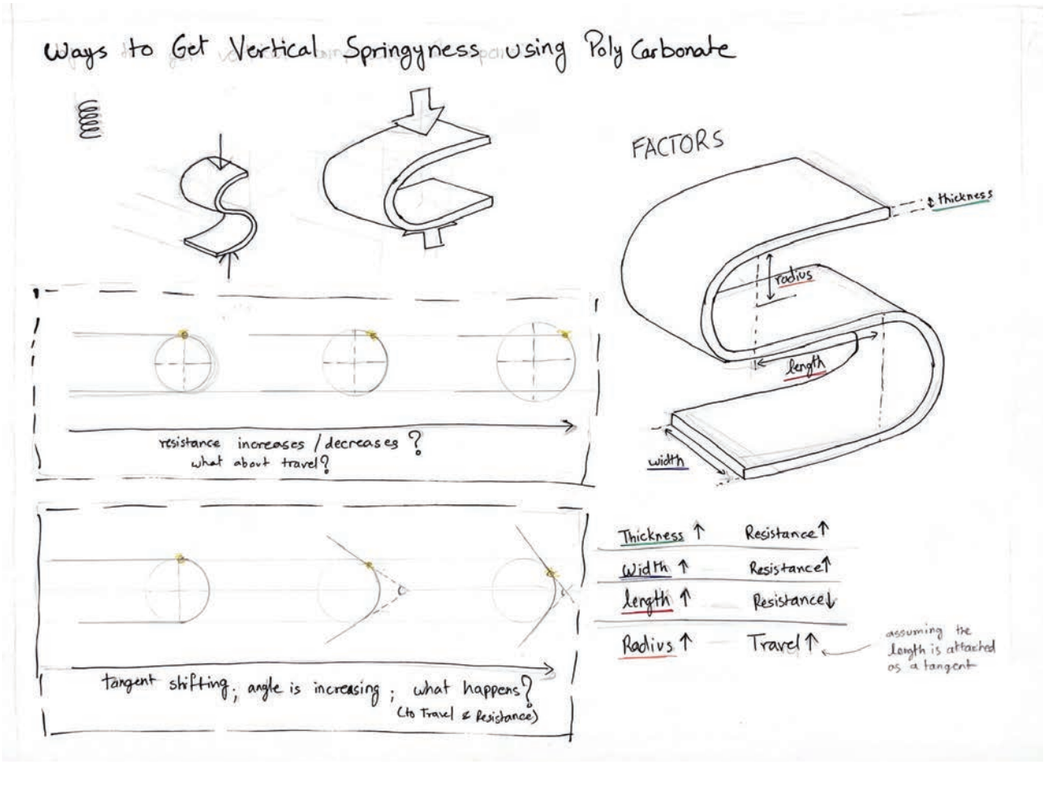


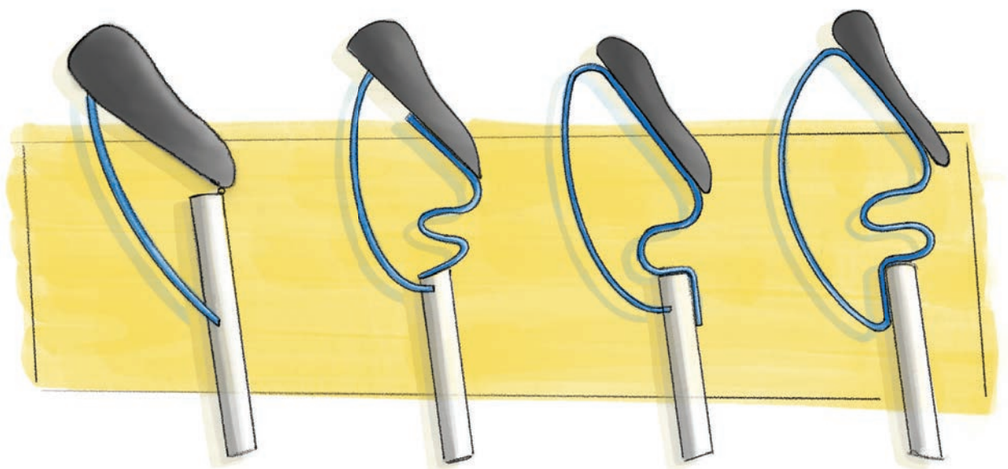


Testing the compliant mechanism

Development

Developing the form of the compliant mechanism using polycarbonate





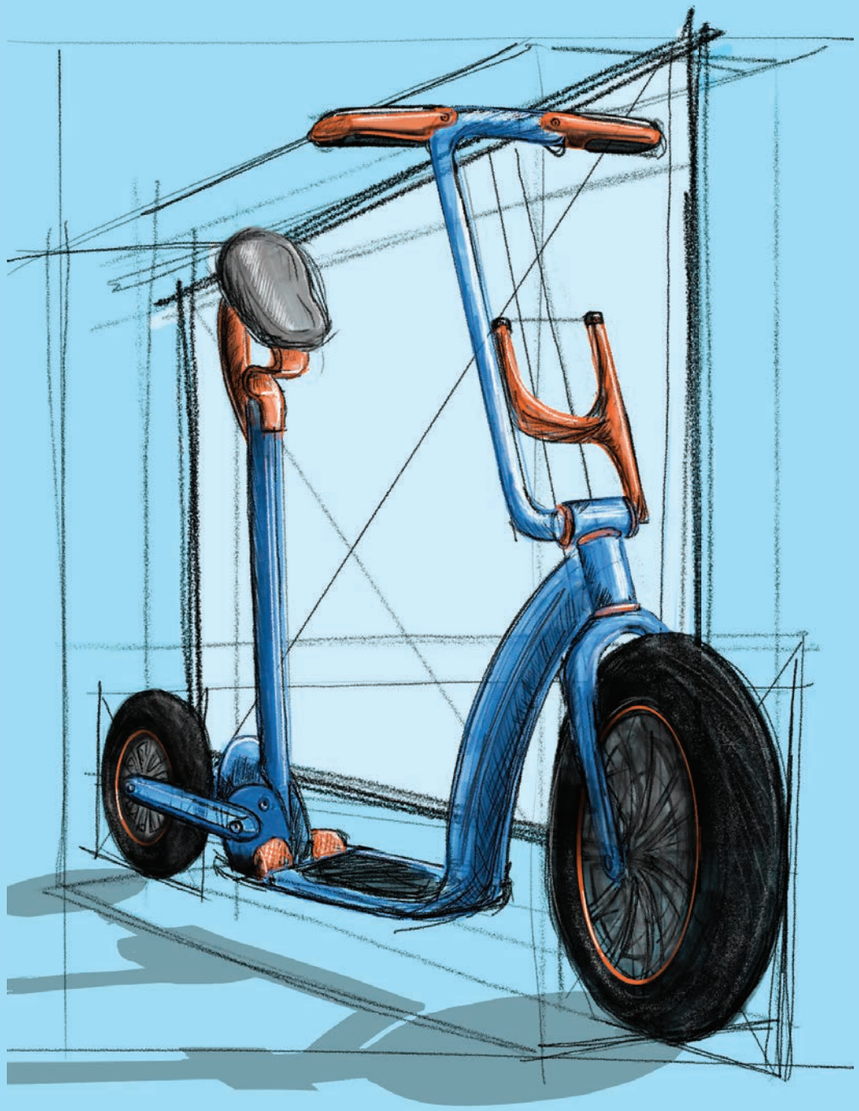
Prototype

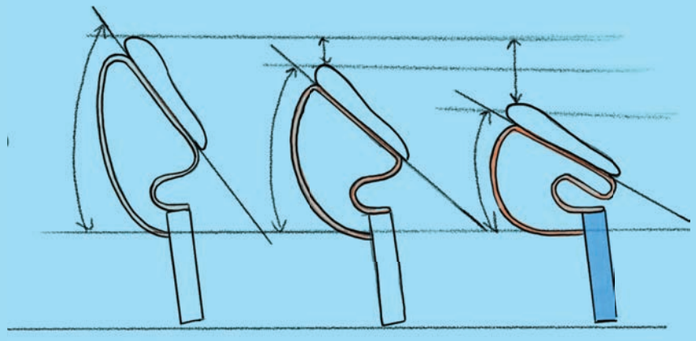
Developing the form of the compliant mechanism using polycarbonate



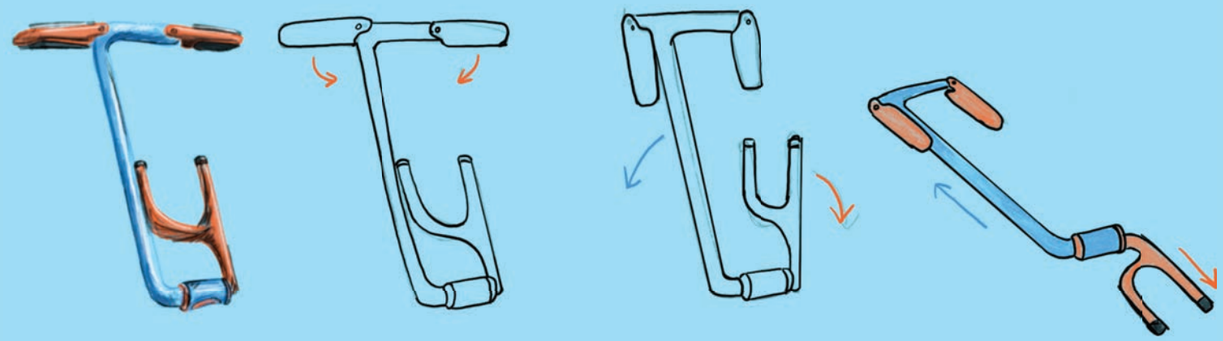


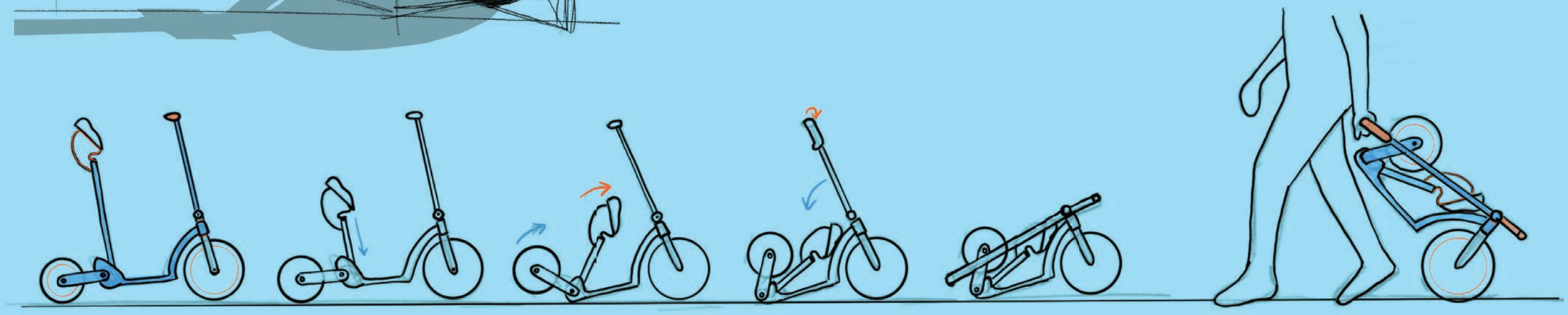
Renders











Renders



